

My Home Path: Leveraging Game Design to Foster Bicultural Identity and Intergenerational Connection in Immigrant Communities

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Figure 1: My Home Path Dining Room Scene

Abstract

Immigrant children face unique challenges, including dual-language development, cultural differences, and acculturation, which shape their mental health, academic performance, and social integration. Proficient bilingualism and bicultural competence are essential for enhancing their well-being while promoting empathy, cultural pride, and meaningful intergenerational dialogue. *My Home Path* is

designed to explore how game design can serve as a transformative tool for second-generation Chinese immigrant children, enabling them to navigate and embrace their bicultural identity. By immersing players in the first-person experiences of their parents' immigrant journeys, the game leverages bilingual storytelling and interactive cultural objects to foster empathy and a deeper understanding of heritage. Players engage with narratives rooted in childhood experiences, memories, and anecdotes from first-generation immigrants, supporting cultural connection and belonging. Through this game, we hope to investigate the potential of games for cultural education, preserving heritage, and promoting intergenerational dialogue within immigrant communities.

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CCS Concepts

• **Software and its engineering** → **Interactive games.**

Keywords

Game Design, Intergenerational Storytelling, Transformational Game, Interactive Storytelling, Gaminiscing, Composite Narratives, Bicultural Identity, Bilingualism

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1 Introduction

Immigrant families often face socioeconomic disadvantages that negatively impact their children's academic performance and social integration [19]. Immigrant children also encounter challenges such as dual language development, cultural differences, acculturation, and adaptation, all of which shape their mental health outcomes [20]. There is a growing need to support the assimilation of immigrant children in a way that helps them thrive in American society while preserving their cultural identities. To support the well-being of immigrant children, fostering cultural adaptation and social integration is crucial. Developing bicultural competence is considered the ideal outcome for acculturation. In addition, proficient bilingualism offers academic, cognitive, and social advantages [20]. Together, dual-language and bicultural competence significantly enhance the overall well-being of immigrant children.

Moreover, intergenerational storytelling also nurtures social connection within families and plays a unique role in immigrant communities, where it helps navigate cultural and language differences, preserve historical legacies, and shape family identity [1, 13, 17]. Games serve as powerful platforms to support immigrant children in enhancing their bilingualism and bicultural competence through storytelling. Educational games uniquely cultivate engagement, motivation, and knowledge retention by integrating active and experiential learning [2]. By creating shared, interactive spaces, games facilitate intergenerational conversations and help bridge cultural and language barriers through intuitive storytelling. Incorporating interactive storytelling elements enables players to explore and preserve cultural and personal histories, making the experience both meaningful and educational [16]. Using the Chinese immigrant experience as an example, we designed an interactive narrative-driven game based on composite narratives [12], *My Home Path*, to explore how a transformational game [4] can support intergenerational storytelling. This approach aims to help second-generation Chinese immigrant children enhance their bilingualism and bicultural competence while strengthening cultural pride, and a deeper understanding of their heritage. It also provides a meaningful platform for a broader audience seeking cultural connection and cross-cultural empathy.

2 Related Work

Immigration-themed video games allow players to experience the challenges of immigration and foster empathy by immersing them in personal and cultural narratives. Games like *Bury Me, My Love* [18], *City of Immigrants* [7], and *Catch the Butterfly* [6] exemplify this approach by highlighting the struggles, resilience, and family-driven motivations of immigrants through interactive storytelling. *Bury Me, My Love* and *City of Immigrants* are set against historically and socially significant backdrops, where players navigate decision-making processes that impact character relationships and outcomes. Similarly, *Catch the Butterfly* shares the real and personal journey of an immigrant facing job loss, financial instability, and the loss of family and friends, emphasizing resilience through immersive storytelling. Our game draws significant inspiration from *Catch the Butterfly*'s storytelling approach and the concept of "gaminiscing", defined as "using game technology to archive and recreate personal oral histories as an engaging experience" [6]. Gaminiscing is exemplified by *Brukel*, a 3D recreation of the designer's grandmother's home while hearing her firsthand accounts of surviving World War II [5]. These games and concept elements inspired our initial design approach, emphasizing the preservation and sharing of composite personal stories through immersive storytelling gameplay.

In addition to being inspired by immigration-themed games, storytelling and gaminiscing concept, narrative interaction plays a crucial role in the game by creating a sense of agency and environmental storytelling [10, 15]. Games like *To the Moon* [8] and *What Remains of Edith Finch* [9], which focus on interactive storytelling and memories with simple exploration, served as the initial inspiration for the design mechanics of our game. Then, games like *Valiant Hearts* [21] and *1979 Revolution: Black Friday* [11] emphasize storylines and player agency, allowing players to make choices that shape character relationships, plot outcomes, and the emotional tone of the story. These mechanics create deeply immersive and personalized narrative experiences. *Valiant Hearts*, a puzzle-adventure game that follows the intertwined stories of four characters during World War I, blends hand-drawn comic-style visuals, environmental puzzles, and narrative-driven gameplay to create a poignant, educational experience. Its use of interactivity and historical artifacts inspired us to further explore hand-drawn aesthetics and interactive artifacts in our own game design. Our game also draws particular design inspiration from *1979 Revolution: Black Friday*, which incorporates real historical events and photographs, with the protagonist's experiences grounded in true historical contexts. This approach effectively connects players to the story while leaving a lasting impression of the historical events. We designed our game by combining elements of immigrant storytelling, gaminiscing and interactive visual engagement. Players will not only learn about cultural heritage and personal histories but also provide the opportunity to feel a deeper emotional connection from the game art and narratives. Ensure intergenerational storytelling is both educational and impactful through interactive design.

3 Game Design

My Home Path is a single-player interactive narrative game designed to bridge generations by immersing second-generation Chinese immigrant children in the experiences of their parents' immigration

journeys through engaging object interactions. The game incorporates the concept of gaminiscing and interactive storytelling, combining composite narratives, interactive design, and a traditional Chinese art style (see Figure 2) to create an engaging and immersive experience centered around Chinese cultural themes. By incorporating simple click-based interactions, this digital game enables children and their grandparents to connect and engage in meaningful play and conversations. Our hope is that by making the game available for free online that family members across generations can easily share the game with one another.

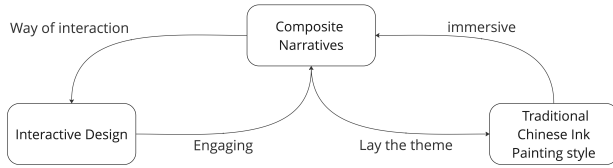


Figure 2: Game Design Flow Chart

3.1 Game Interaction Design

We focused on designing diverse interactions with game objects to make the interactive narrative more intuitive, meaningful, and educational. We carefully selected 14 objects that embody significant aspects of Chinese cultural heritage, values, symbolism and family memory based on the design framework for reflective play [14]. To initially brainstorm objects, we used a combination of informal interviews with Chinese immigrant families as well as the personally meaningful experiences of each of the four co-authors (each of whom are Chinese students studying in the U.S. and relatedly have bicultural experiences). The final 14 objects were then chosen to reflect meaningful traditions and family experiences passed down through generations. The storytelling was then developed using composite narratives, blending insights from Chinese family interviews to align with key cultural themes, such as mindfulness, mentorship, and resilience. This approach ensures that each interaction provides players with opportunities to explore and learn about their heritage, fostering a deeper connection to their cultural identity. We added minigames, animations, audio and text interactions that bring these items to life, providing an engaging and immersive way for players to interact with the game and also teach about cultural significance. **MiniGame:** The Go minigame (see Figure 3) introduces the basic rules of the game while highlighting its role in developing mentorship between older and younger Chinese individuals and for the older generation to reminisce about the past. **Animation:** The Chinese orchid watering animation is accompanied by a story about a grandmother’s daily habits and the cultural symbolism of orchids in Chinese heritage, representing elegance and resilience. The calligraphy animation (see Figure 4) not only demonstrates the proper technique for writing the Chinese character “family” but also conveys the deeper meaning of calligraphy as a practice that cultivates mindfulness. **Audio Interaction:** the Erhu plays Butterfly Lovers with a single click, its tone and melody evoking deep emotions and nostalgia tied to home and family. **Text Interaction:** the letters serve as a record of the father’s

early studying abroad experiences, offering a personal connection to the immigrant journey.

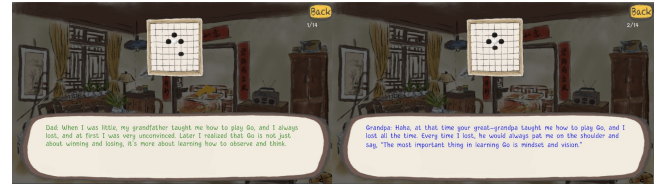


Figure 3: Minigame: Show the basic rules of Chinese board game Go



Figure 4: Animation: Demonstrate Chinese calligraphy by using brush copy writing the Chinese character “family”

3.2 Game Art and Narrative Design

Game art and narrative design are central to creating an immersive and nostalgic experience for players. Our game adopts the traditional Chinese Ink Painting style that prioritizes poetic meaning and artistic intention over realistic representation (see Figure 5). This art style serves as a powerful tool to depict the living space and objects within the game, transforming them into evocative elements that convey cultural significance and emotional depth. The stories in this game are crafted from composite narratives based on interviews with multiple Chinese immigrants and our producers’ personal experiences. To convey the story and significance of each interactive object, a dialogic narrative unfolds every time the player clicks on an object. Each unique item reveals a dialogue, accompanied by text, that explores its connection to specific family members and its role within the family dynamic, like the Chinese tea culture passed from grandpa to dad. Through these interactions, players uncover the philosophical and cultural meaning behind the objects. We also incorporated Chinese voice overs for all the stories, providing a bilingual learning experience as well as enriches the storytelling.



Figure 5: Traditional Chinese ink painting style depicted dining room and bedroom scenes

4 Playtesting

We conducted six initial playtesting [3] studies with both general players without Chinese immigration background and players with second-generation Chinese immigrant backgrounds (see Figure 6). Early two playtests focused on the game’s aesthetics and mechanics, with participants highlighting minigames such as finding a lucky coin in dumplings and mixing and matching Spring Festival scrolls as particularly engaging and memorable. Feedback from these sessions emphasized the need to improve interactivity by incorporating more clickable objects with animations and refining narrative delivery using speech bubbles and better pacing. This feedback guided iterative development to enrich the game’s cultural depth and interactivity.

As the prototypes evolved, later playtests shifted toward examining the cultural impact and transformative experiences of players. Players from second-generation Chinese backgrounds noted how the game evoked family memories, increased their awareness of Chinese traditions, and strengthened generational connections. Interactive elements such as playing Go, and participating in a tea ceremony, exploring the meaning behind everyday objects were especially effective in enhancing cultural understanding. Based on feedback, we further improved animation to align with storytelling and incorporated multimodal interactions, including music, audio, and text, to create a more engaging and accessible experience.

All the playtests revealed the game’s educational value and transformative potential. One of the Chinese background players reported increased confidence in discussing cultural traditions with their families and a deeper appreciation for Chinese heritage. One of the second-generation Chinese background players noted that listening to background stories and reading bilingual letters allowed them to practice their Chinese while learning about their heritage. Some players expressed excitement at discovering familiar traditions, such as a shared experience of learning Go from their grandparents, while others were intrigued to learn something new, such as the symbolic meaning of the Chinese orchid. Surprisingly, general players also resonated with elements like the music played on the Erhu, the family album, and the grandmother’s sweater, finding parallels to their own family gatherings and traditions. These findings suggest that the game has educational and transformational potential for cultural understanding and connection across diverse audiences.

5 Critical Reflection

My Home Path game addresses the challenges faced by second-generation immigrant communities, including disconnection from cultural heritage, understanding bicultural identity, and developing dual-language competence. By integrating personal experiences with culturally significant objects, the game provides an opportunity for players to deepen their understanding of cultural identity, heritage, and bilingual learning. Through interactive storytelling and gamifying elements, culturally rich gameplay, and interactive visual engagement, players are invited to explore the challenges and triumphs inherent in their heritage, discovering connections between these narratives and their own bicultural experiences. This reflective journey empowered a sense of cultural identity, enabling players to appreciate the significance of their roots.

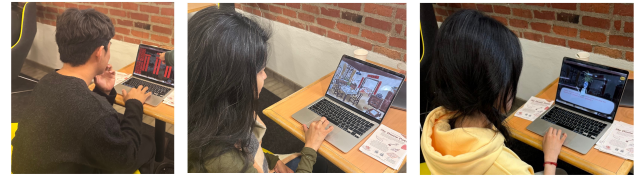


Figure 6: Playtests with different Players

The gameplay experience emphasizes intergenerational conversations and language learning, helping players connect more deeply with their family histories. By interacting with culturally significant objects and uncovering embedded stories, players engage with their heritage through both memory and dialogue. The game encourages players to explore meaningful family traditions and reflect on how these practices have shaped their identities. As they navigate the narratives, players learn culturally specific knowledge, like the wisdom of Go, fostering a more intimate and interactive connection to their cultural heritage.

Early playtest findings indicate that as players progress through the game, their emotional responses and perspectives evolve, reflecting the shifting feelings toward their Chinese heritage. Specifically, gameplay sparked moments of connection to personal histories and players frequently unsolicited shared stories of their families as they came across new objects in the game. This emotional connection cultivate empathy across generations, highlighting the game’s potential to resonate with a broader immigrant audience to connect with those personal stories and learn from those cultural traditions.

6 Future Iteration

This game prototype demonstrates the concept of using games as a platform to foster intergenerational dialogue and support cultural connection, bilingualism, and bicultural competence among second-generation Chinese immigrant children. Our next steps are to refine each minigame and enhance the interactive experience to encourage concurrently collaborative conversations through gameplay. Also, conducting more formal study on the emotional and identity-relevant impacts of the game (e.g. whether narrative elements and interactive design guide players from feelings of distance or uncertainty to a deeper sense of connection and pride in their bicultural identity). Additionally, we envision integrating AI tools to customize the game’s environment and objects, tailoring them to the unique memories and stories of diverse immigrant audiences, further expanding *My Home Path*’s potential to connect with and support various individual immigrant and cultural communities.

7 Conclusion

My Home Path game highlights the transformative potential of interactive storytelling in fostering cultural connection, bilingualism, and bicultural competence among second-generation Chinese immigrant children and supporting their overall well being. By integrating culturally significant and interactive objects, bilingual voice overs and composite narratives, the game serves as a dynamic platform for bridging generational gaps and encouraging meaningful dialogue about Chinese heritage. Playtesting studies

demonstrate the game's ability to evoke family memories, enhance Chinese cultural understanding, and inspire players to embrace their bicultural identities. It also underscores the broader potential of games as transformational tools for education, empathy, and intergenerational storytelling across diverse immigrant communities. Moving forward, we envision conducting more formal study and integrating AI tools to enhance game quality and experience for diverse cultural communities and individuals.

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